KiPI AC impedance report: 12V

Audio_Visualiser_DSP.kicad_pcb · 2026-10-19 23:07 · KiPI 0.9.2 · solved in 367.7 s

Rail **12V**: +12V, +12Vin, Net-(U201-OUT) against GND, from 1 kHz to 300 MHz. 7 capacitor(s) with models, 2 series link(s); plane capacitance 0.578 nF.

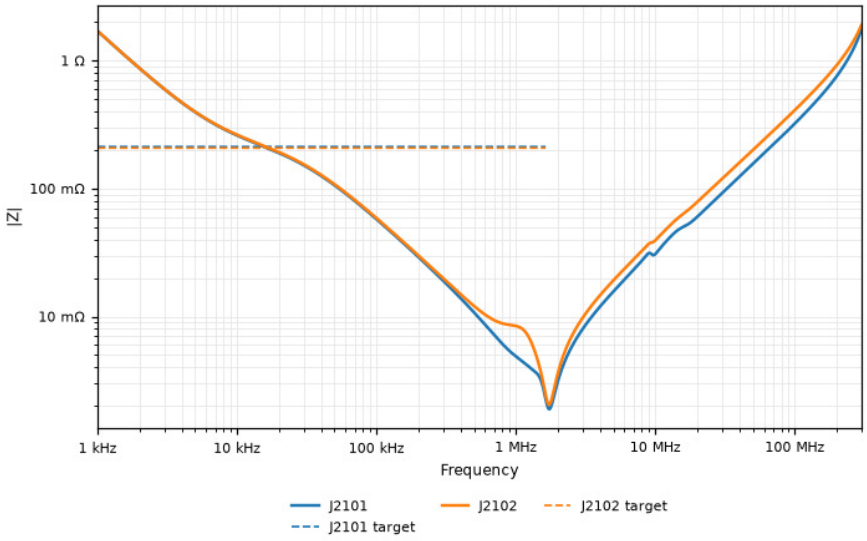
Overall: **FAIL** (0 of 2 probe(s) with a target pass).

Probes

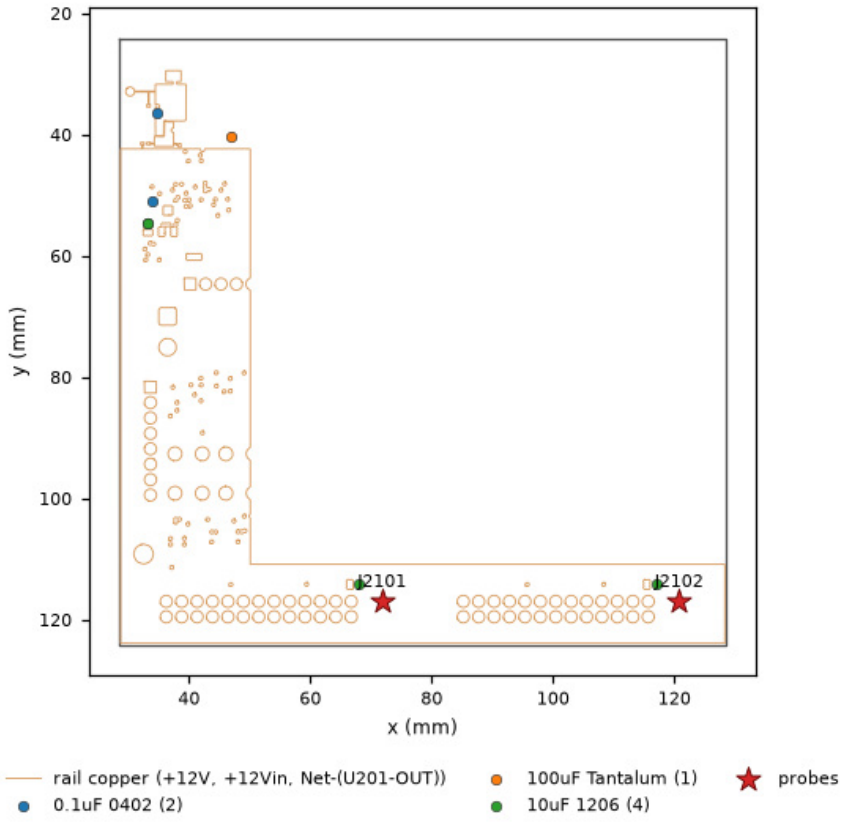
Probe	Pins (rail / ground)	Min |Z|	Worst |Z|	at	Target	Margin	Result
J2101	J2101 / J2101	1.89 mΩ	1.703 Ω	1 kHz	212.9 mΩ to 1.637 MHz	-18.06 dB	**FAIL**
J2102	J2102 / J2102	2.047 mΩ	1.704 Ω	1 kHz	207.7 mΩ to 1.637 MHz	-18.28 dB	**FAIL**

Worst |Z|: where the impedance comes closest to (or furthest over) the probe's target, up to where the target applies; the highest impedance in the sweep without one. Margin: target over |Z| there (positive passes).

Impedance



Board map



Seen from the top, with the rail's copper on every layer merged.

Capacitor models

Model	C	C derated	ESL	ESR	Self-resonance	Parts on rail	References
0.1uF 0402	100 nF	59.67 nF to 60.14 nF at 11.8 V to 12 V (60% to 60%)	200 pH	20 mΩ	35.59 MHz	2	C202, C405
100uF Tantalum	100 µF	67.16 µF at 11.8 V (67%)	2.5 nH	300 mΩ	318.3 kHz	1	C203
10uF 1206	10 µF	6.716 µF to 6.727 µF at 11.8 V to 11.8 V (67% to 67%)	1.2 nH	2 mΩ	1.453 MHz	4	C402, C403, C2101, C2102

Capacitors derated: 7 for DC bias at the voltages the last DC solve found across them. They keep 60% to 67% of their capacitance. DC bias follows a typical class II ceramic's curve through each model's capacitance kept at its voltage rating; aging takes each model's rate per decade of hours past 1000 hours; a manufacturing tolerance is applied at its low end.

94.16 µF of capacitors in the model, plus 0.578 nF between the rail's copper and GND. ESL is the part's own; the mounting (pads, vias, planes) comes from the field solve.

Model

Series links

Link	Side A	Side B	R	ESL
U201	U201 · +12Vin	U201 · Net-(U201-OUT)	28.2 mΩ	
R207	R207 · Net-(U201-OUT)	R207 · +12V	10 mΩ	

Sweep and mesh

Setting	Value
Sweep	1 kHz to 300 MHz, 1370 points (250 per decade)
Field solves	111 (20 per decade; the copper's response is interpolated in between)

Setting	Value
Mesh	max edge 0.5 mm, min edge 0.03 mm at plated holes, grading 0.3, only at holes near the rail
Dielectric	loss tangent 0.02; εr top to bottom: 3313 RC57% 99 µm εr 4.1, FR4 core 100 µm εr 4.6, 2116 RC54% x3 357 µm εr 4.16, FR4 core 300 µm εr 4.6, 2116 RC54% x3 357 µm εr 4.16, FR4 core 100 µm εr 4.6, 3313 RC57% 99 µm εr 4.1
Copper	resistivity 1.724e-08 Ω·m
Via plating	micro any drill: 25 um, if filled: non-conductive epoxy; buried any drill: 25 um, if filled: non-conductive epoxy; blind any drill: 25 um, if filled: non-conductive epoxy; blind_buried any drill: 25 um, if filled: non-conductive epoxy; through any drill: 25 um, if filled: non-conductive epoxy; pth any drill: 25 um, if filled: non-conductive epoxy; any any drill: 25 um, if filled: non-conductive epoxy (first matching rule)
Temperature	not modeled: copper at 20 °C, capacitors at their rated values

Solver

Quantity	Value
Mesh	793635 nodes, 1443379 triangles
System	1044455 unknowns, 14130903 non-zeros, 11 ports
Direct solver	pardiso
Time	367.7 s (setup 34.0 s, field 333.6 s)

Notes

- Series link FB702 leads off the rail to GNDA_DSP; left out

User notes